The Yield Recurrence

May 13, 2026

The recurrence we consider in this section is the following.

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1 \right) + J^{(n)} \max \left(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n} \right), \forall n \geq 1, \quad Y_1 = Y_2 = 0. \tag{1.1}$$

Let us explain the notations. Let Y_n be the yield of the tree with n nodes. The variables I and J are Bernoulli with expectation u and v respectively, such that $I + J \leq 1$. That means we can only either take the sum-plus-one term, or the max term, or zero. The variable U_n is uniformly distributed on $\{1, \dots, n\}$. Superscripts are for independent copies. We assume that all collections of copies are independent. Now we can understand why it is correct to reuse the same variables $Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n}$ and U_n across both terms.

1.1 Lower and Upper Bounds of the Expectation

Let $y_n = \mathbb{E}[Y_n]$. Take expectation on both sides, we have

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^{n} y_i + \frac{v}{n} \sum_{i=1}^{n} \mathbb{E}[\max(Y_i, Y_{n+1-i})].$$

For an upper bound, using $\max(Y_i, Y_{n+1-i}) \leq Y_i + Y_{n+1-i}$, we get an upper bound sequence

$$y_{n+2} = u + \frac{2(u+v)}{n} \sum_{i=1}^{n} y_i, \quad y_1 = y_2 = 0. \tag{1.2}$$

There are several ways to get a lower bound. The first one is to use convexity of the absolute value function.

$$\begin{aligned} \mathbb{E}[\max(Y_i, Y_{n+1-i})] &= \mathbb{E}\left[\frac{1}{2}(Y_i + Y_{n+1-i} + |Y_i - Y_{n+1-i}|) \right] \\ &\geq \frac{1}{2}(\mathbb{E}[Y_i] + \mathbb{E}[Y_{n+1-i}]) + \frac{1}{2}|\mathbb{E}[Y_i] - \mathbb{E}[Y_{n+1-i}]| \\ &= \frac{1}{2}(y_i + y_{n+1-i} + |y_i - y_{n+1-i}|). \end{aligned}$$

This gives

$$y_{n+2} = u + \frac{2u+v}{n} \sum_{i=1}^{n} y_i + \frac{v}{2n} \sum_{i=1}^{n} |y_i - y_{n+1-i}|, \quad y_1 = y_2 = 0. \tag{1.3}$$

Another way is to use convexity of the max function.

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i,n+1-i)}.$$

This gives

$$y_{n+2} = u + \frac{2u}{n}\sum_{i=1}^{n} y_i + \frac{v}{n}\sum_{i=1}^{n} y_{\max(i,n+1-i)}, \quad y_1 = y_2 = 0. \tag{1.4}$$

If we can prove that y_n is non-decreasing in n, then we can ignore the middle term in the sum of maxima, and get

$$y_{n+2} = u + \frac{2u}{n}\sum_{i=1}^{n} y_i + \frac{2v}{n}\sum_{i=\lfloor (n+1)/2 \rfloor}^{n} y_i, \quad y_1 = y_2 = 0. \tag{1.5}$$

The weakest lower bound is to replace the max by the average, which gives

$$y_{n+2} = u + \frac{2u+v}{n}\sum_{i=1}^{n} y_i, \quad y_1 = y_2 = 0. \tag{1.6}$$

When $2u+v>1$, the true expectation is bounded between by polynomial of degrees $2u+v-1$ and $2(u+v)-1$. This case agrees with real data. We conjecture that the true expectation also grows polynomially. Details of the analysis of the deterministic recurrences are in Chapter **??**.

1.2 Combinatorial Analysis

For convenience, let us shift the index. The recurrence becomes

$$Y_{n+1} \stackrel{d}{=} I^{(n)}\left(Y^{(1)}_{U_n} + Y^{(2)}_{n-1-U_n} + 1\right) + J^{(n)} \max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n-1-U_n}\right), \forall n \geq 1, \quad Y_0 = Y_1 = 0. \tag{1.7}$$

The variable U_n is now distributed on $\{0,\dots,n-1\}$. Let $p_{n,m} = \mathbb{P}(Y_n = m)$ and $q_{n,m} = \mathbb{P}(Y_n \leq m)$. We have the relations

$$p_{n,m} = q_{n,m} - q_{n,m-1} \text{ and } q_{n,m} = \sum_{i=0}^{n-1} p_{i,m},$$

and initial conditions

$$p_{0,0} = p_{1,0} = 1 \text{ and } q_{0,m} = q_{1,m} = 1 \text{ for all } m \geq 0.$$

From the recurrence, we have

$$\begin{aligned}
&nq_{n+1,m} \\
&= nu\mathbb{P}\left(Y^{(1)}_{U_n} + Y^{(2)}_{n-1-U_n} + 1 \leq m\right) + nv\mathbb{P}\left(\max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n-1-U_n}\right) \leq m\right) + n(1-u-v) \\
&= u\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} p_{i,j}q_{n-1-i,m-1-j} + v\sum_{i=0}^{n-1} q_{i,m}q_{n-1-i,m} + n(1-u-v) \\
&= u\left(\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} q_{i,j}q_{n-1-i,m-1-j} - \sum_{i=0}^{n-1}\sum_{j=0}^{m-2} q_{i,j}q_{n-1-i,m-2-j}\right) \\
&\quad + v\sum_{i=0}^{n-1} q_{i,m}q_{n-1-i,m} + n(1-u-v).
\end{aligned}$$

Let $Q_m(z) = \sum\limits_{n=0}^{\infty} q_{n,m} z^n$ be the generating function of $q_{n,m}$ when m is fixed. Multiplying both sides by z^{n-1} and summing over n, we get

$$\sum_{n=0}^{\infty} n\, q_{n+1,m} z^{n-1} = \frac{d}{dz} \sum_{n=0}^{\infty} q_{n+1,m} z^n = \frac{d}{dz} \left(\frac{Q_m(z) - 1}{z} \right) = \frac{z Q'_m(z) - Q_m(z) + 1}{z^2},$$

$$\sum_{n=0}^{\infty} \sum_{i=0}^{n-1} \sum_{j=0}^{m-1} q_{i,j} q_{n-1-i,m-1-j} z^{n-1} = \sum_{j=0}^{m-1} Q_j(z) Q_{m-1-j}(z),$$

$$\sum_{n=0}^{\infty} \sum_{i=0}^{n-1} \sum_{j=0}^{m-2} q_{i,j} q_{n-1-i,m-2-j} z^{n-1} = \sum_{j=0}^{m-2} Q_j(z) Q_{m-2-j}(z),$$

$$\sum_{n=0}^{\infty} \sum_{i=0}^{n-1} q_{i,m} q_{n-1-i,m} z^{n-1} = Q_m(z)^2,$$

$$\sum_{n=1}^{\infty} n(1-u-v) z^{n-1} = \frac{1-u-v}{(1-z)^2}.$$

We get the following functional equation for $Q_m(z)$.

$$z Q'_m(z) - Q_m(z) + 1 = z^2 \left(u \sum_{j=0}^{m-1} Q_j(z) Q_{m-1-j}(z) - u \sum_{j=0}^{m-2} Q_j(z) Q_{m-2-j}(z) + v Q_m(z)^2 \right) + \frac{(1-u-v) z^2}{(1-z)^2}. \tag{1.8}$$

In particular, when $m = 0$, we have

$$z Q'_0(z) - Q_0(z) + 1 = v z^2 Q_0(z)^2 + \frac{(1-u-v) z^2}{(1-z)^2}. \tag{1.9}$$

By the convention of Riccati equations, let $Q_0(z) = -\frac{1}{vz} \frac{W'(z)}{W(z)}$, we have

$$z \left(\frac{1}{vz^2} \frac{W'(z)}{W(z)} - \frac{1}{vz} \frac{W''(z) W(z) - W'(z)^2}{W(z)^2} \right) + \frac{1}{vz} \frac{W'(z)}{W(z)} + 1 = \frac{1}{v} \frac{W'(z)^2}{W(z)^2} + \frac{(1-u-v) z^2}{(1-z)^2}.$$

Equivalently,

$$W''(z) - \frac{2}{z} W'(z) - v \left(1 - \frac{(1-u-v) z^2}{(1-z)^2} \right) W(z) = 0. \tag{1.10}$$

Our approach follows closely [1].

Step 1. By Cauchy-Hadamard theorem, the radius of convergence of $Q_0(z)$ is larger than or equal to 1. Suppose that it is strictly larger than 1. Then $Q_0(z), Q'_0(z)$ is finite. Taking the limit of both sides of (1.9) as $z \to 1^-$, we get a contradiction. Therefore, the radius of convergence of $Q_0(z)$ is exactly 1.

Step 2. We find the indicial equation of the ODE at $z = 1$.

A Tighter Upper Bound on the Yield Recurrence

May 20, 2026

Let us recall the yield recurrence

$$Y_{n+2} \overset{d}{=} I^{(n)} \left(Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1\right) + J^{(n)} \max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n}\right), \forall n \geq 1, \quad Y_1 = Y_2 = 0.$$

2.1 Escaping from the Random Variables

Let $Z_n = \lambda^{Y_n}$, where $\lambda > 0$ is a constant to be determined later. For every $I \in \{0, 1\}$ and $C \in \mathbb{R}$, we have

$$C^I = I \cdot C + (1 - I).$$

Since $I^{(n)} J^{(n)} = 0$, we have

$$\begin{aligned}
&\lambda^{Y_{n+2}} \\
&= \left(I^{(n)}\left(\lambda^{Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1}\right) + 1 - I^{(n)}\right)\left(J^{(n)}\left(\lambda^{\max(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n})}\right) + 1 - J^{(n)}\right) \\
&= I^{(n)}\left(\lambda^{Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1}\right) + J^{(n)}\left(\lambda^{\max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n}\right)}\right) + 1 - J^{(n)} - I^{(n)} \\
&\leq I^{(n)}\left(\lambda^{Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1}\right) + J^{(n)}\left(\lambda^{Y^{(1)}_{U_n}} + \lambda^{Y^{(2)}_{n+1-U_n}}\right) + 1 - I^{(n)} - J^{(n)}
\end{aligned}$$

Let $z_n = \mathbb{E}[Z_n]$. Taking the expectation, we get

$$z_{n+2} \leq \frac{\lambda u}{n} \sum_{i=1}^{n} z_i z_{n+1-i} + \frac{2v}{n} \sum_{i=1}^{n} z_i + 1 - u - v.$$

Hence let us consider the sequence

$$n z_{n+2} = \lambda u \sum_{i=1}^{n} z_i z_{n+1-i} + 2v \sum_{i=1}^{n} z_i + n(1 - u - v), \forall n \geq 1, \quad z_1 = z_2 = 1. \tag{2.1}$$

Experiments show that (z_n) grows exponentially, the bound fails. That means, using $\max(x, y) \leq x + y$ keeps the sequence sublinear, but using $\lambda^{\max(x,y)} \leq \lambda^x + \lambda^y$ pushes it to linear.

2.2 Using Techniques from Analytic Combinatorics

Let $Z(t) = \sum_{n=1}^{\infty} z_n t^n$ be the generating function for the sequence $\{z_n\}$. Let us analyze each term in (2.1) through Z.

$$\begin{aligned}
&\sum_{n=1}^{\infty} n z_{n+2} t^n = t\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Z(t) - z_1 t - z_2 t^2}{t^2}\right) = t\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Z(t)}{t^2} - \frac{1}{t} - 1\right) = \frac{Z'(t)}{t} - \frac{2Z(t)}{t^2} + \frac{1}{t},\\
&\sum_{n=1}^{\infty}\sum_{i=1}^{n} z_i z_{n+1-i} t^n = \frac{Z(t)^2}{t},\\
&\sum_{n=1}^{\infty}\sum_{i=1}^{n} z_i t^n = \frac{Z(t)}{1-t},\\
&\sum_{n=1}^{\infty} n t^n = \frac{t}{(1-t)^2}.
\end{aligned}$$

The function equation is

$$\frac{Z'(t)}{t} - \frac{2Z(t)}{t^2} + \frac{1}{t} = \lambda u\frac{Z(t)^2}{t} + 2v\frac{Z(t)}{1-t} + (1-u-v)\frac{t}{(1-t)^2}.$$

Equivalently,

$$Z'(t) = \lambda u Z(t)^2 + \left(\frac{2}{t} + \frac{2vt}{1-t}\right) Z(t) + \frac{(1-u-v)t^2}{(1-t)^2} - 1. \tag{2.2}$$

Substituting $Z(t) = -\dfrac{1}{\lambda u}\dfrac{W'(t)}{W(t)}$ in the convention of a Riccati equation, we get

$$\begin{aligned}
&-\frac{1}{\lambda u}\frac{W''(t)W(t) - W'(t)^2}{W(t)^2} = \frac{1}{\lambda u}\frac{W'(t)^2}{W(t)^2} - \frac{1}{\lambda u}\left(\frac{2}{t} + \frac{2vt}{1-t}\right)\frac{W'(t)}{W(t)} + \frac{(1-u-v)t^2}{(1-t)^2} - 1\\
\iff\ &-\frac{1}{\lambda u}\frac{W''(t)}{W(t)} = -\frac{1}{\lambda u}\left(\frac{2}{t} + \frac{2vt}{1-t}\right)\frac{W'(t)}{W(t)} + \frac{(1-u-v)t^2}{(1-t)^2} - 1\\
\iff\ &W''(t) - \left(\frac{2}{t} + \frac{2vt}{1-t}\right)W'(t) + \left(\frac{(1-u-v)t^2}{(1-t)^2} - 1\right)\lambda u W(t) = 0.
\end{aligned}$$

References

[1] Florent Koechlin and Pablo Rotondo. “The effects of semantic simplifications on random BST-like expression-trees”. In: *Discrete Mathematics* 347.5 (2024), p. 113906.